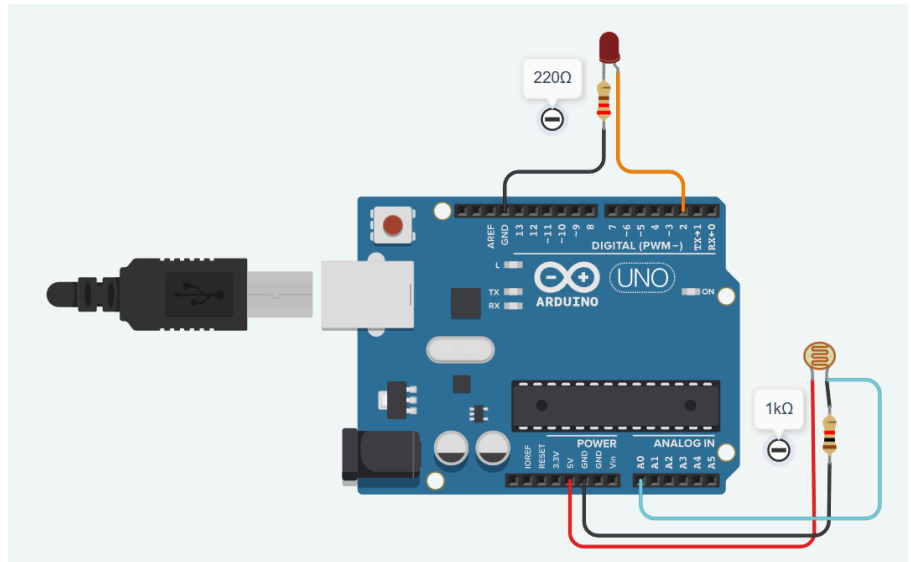


q1) Write an algorithm/flowchart and C++ program for a smart street light system with the following requirements:

```

1  int LED = 2;
2  int LDR = A0;
3
4  void setup()
5  {
6      Serial.begin(9600);
7      pinMode(LED, OUTPUT);
8      pinMode(LDR, INPUT);
9  }
10
11 void loop()
12 {
13     int LDR_value = analogRead(LDR);
14     Serial.println(LDR_value);
15     if(LDR_value <= 300){
16         digitalWrite(LED, HIGH);
17     }
18     else{
19         digitalWrite(LED, LOW);
20     }
21 }

```



q2)An IoT system contains 4 sensor nodes arranged in rows and columns. Using nested loops, print:

```

Sensor(1,1) Sensor(1,2) Sensor(1,3)
Sensor(2,1) Sensor(2,2) Sensor(2,3)
Sensor(3,1) Sensor(3,2) Sensor(3,3)
Sensor(4,1) Sensor(4,2) Sensor(4,3)

```

```

1  void setup()
2  {
3      Serial.begin(9600);
4      for(int i = 0; i < 4; i++){
5          for(int j = 0; j < 3; j++){
6              Serial.print("Sensor(");
7              Serial.print(i+1);
8              Serial.print(",");
9              Serial.print(j+1);
10             Serial.print(") ");
11         }
12         Serial.println();
13     }
14 }
15
16
17 void loop()
18 {
19 }

```

q3) note down differences between if-else and switch-case. Where can they be used in IoT?

if-else checks multiple conditions and can handle ranges, logical operators, and complex expressions.

switch-case compares one variable against fixed values and is usually faster and cleaner for many discrete options.

In IoT, if-else can be used for sensor threshold checks (e.g., temperature > 30°C), while switch-case is useful for device modes such as OFF, ON, SLEEP, and ALERT states.

q4) what are system states and how to manage them?

System states represent the current operating condition of a device.

Examples include:

- IDLE
- MONITORING
- PROCESSING
- ERROR.

State management involves tracking the current state using variables, enums, or state machines and defining clear rules for transitions between states. In IoT systems, state management improves reliability, simplifies code organization, and makes device behavior easier to debug and maintain.